Functions & Graphs

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Evaluating functions**

If $f(x) = 3x - 2$, find $f(4)$.

$f(4) = 3(4) - 2 = 10$.

2 **[Foundational]** • **Target: Evaluating functions**

If $f(x) = x^2 + 2x$, find $f(-3)$.

$f(-3) = 9 - 6 = 3$.

3 **[Foundational]** • **Target: Evaluating functions**

Consider $f(x) = 2x^2 - 5x + 1$. Evaluate:

a $f(0)$ 1

b $f(2)$ -1

(a) 1; (b) -1

4 **[Foundational]** • **Target: Domain and range**

The graph of $y = \sqrt{x + 3}$ is shown below. State the domain and range based on the graph, then verify the domain algebraically.

From the graph, the domain is $x \geq -3$ and the range is $y \geq 0$. Algebraically, the expression under the root must be non-negative: $x + 3 \geq 0$, so $x \geq -3$, confirming the graph.

5 **[Foundational]** • **Target: Domain and range**

Find the domain of $f(x) = \frac{1}{x - 5}$.

The denominator cannot equal zero, so $x \neq 5$. Domain: all real numbers except 5.

6 **[Foundational]** • **Target: Domain and range**

Find the domain of $f(x) = \sqrt{2x - 8}$.

$2x - 8 \geq 0$, so $x \geq 4$.

7 **[Foundational]** • **Target: Composition of functions**

If $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $(f \circ g)(2)$.

$g(2) = 4 - 3 = 1$. $f(1) = 2(1) + 1 = 3$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Composition of functions**

If $f(x) = x + 5$ and $g(x) = 3x$, find $(g \circ f)(x)$ as a simplified expression.

$g(f(x)) = 3(x + 5) = 3x + 15$.

9 **[Intermediate]** • **Target: Composition of functions**

If $f(x) = x^2$ and $g(x) = x - 4$, find $(f \circ g)(x)$ and $(g \circ f)(x)$, and state whether they are equal.

$f(g(x)) = (x - 4)^2 = x^2 - 8x + 16$. $g(f(x)) = x^2 - 4$. These are not equal, so function composition is generally not commutative.

10 **[Intermediate]** • **Target: Inverse functions**

Find the inverse of $f(x) = 4x - 8$.

Swap x and y in $y = 4x - 8$: $x = 4y - 8$, so $y = \frac{x+8}{4}$. $f^{-1}(x) = \frac{x+8}{4}$.

11 **[Intermediate]** • **Target: Inverse functions**

Find the inverse of $f(x) = \frac{x-3}{2}$.

Swap: $x = \frac{y-3}{2}$, so $2x = y - 3$, giving $y = 2x + 3$. $f^{-1}(x) = 2x + 3$.

12 **[Intermediate]** • **Target: Inverse functions**

Verify that $f(x) = 5x + 1$ and $g(x) = \frac{x-1}{5}$ are inverses of each other by computing $f(g(x))$.

$f(g(x)) = 5\left(\frac{x-1}{5}\right) + 1 = (x - 1) + 1 = x$, confirming they are inverses.

13 **[Intermediate]** • **Target: Transformations of graphs**

The graph of $y = f(x)$ is shifted 4 units left and 2 units down. Write the new function in terms of f.

$y = f(x + 4) - 2$.

14 **[Intermediate]** • **Target: Transformations of graphs**

If $f(x) = x^2$ and $g(x) = -f(x) + 3$, describe the transformation from f to g.

g reflects f over the x-axis, then shifts the result up 3 units.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Piecewise functions • Cross-Lesson: 4.01 Ratios, Rates & Data

A function is defined by $f(x) = x + 2$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$. Find $f(-3)$ and $f(3)$.

$f(-3) = -3 + 2 = -1$ (using the first piece, since $-3 < 0$). $f(3) = 3^2 = 9$ (using the second piece, since $3 \geq 0$).

16 [Challenging] • Target: Piecewise functions • Cross-Lesson: 5.01 Statistics & Probability

A function is defined by $f(x) = 2x$ for $x \leq 2$ and $f(x) = x + 5$ for $x > 2$. Find $f(2)$ and $f(5)$, and state whether f is continuous at $x = 2$.

$f(2) = 2(2) = 4$ (using the first piece, since $2 \leq 2$). $f(5) = 5 + 5 = 10$. As $x \to 2$ from the right, $f(x) \to 2 + 5 = 7$, which does not match $f(2) = 4$, so f is not continuous at $x = 2$ — there is a jump.

17 [Challenging] • Target: Reading function graphs • Cross-Lesson: 6.01 Geometry

The graph of $f(x) = x^2 - 4$ is shown below. Use the graph to estimate the x-intercepts and y-intercept, then find them exactly.

x-intercepts: solve $x^2 - 4 = 0$, giving $x = 2$ and $x = -2$. y-intercept: $f(0) = -4$.

18 [Challenging] • Target: Even, odd, and other function properties • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Determine whether $f(x) = x^2 - 3$ is even, odd, or neither, using $f(-x)$.

$f(-x) = (-x)^2 - 3 = x^2 - 3 = f(x)$, so f is even.

19 [Challenging] • Target: Even, odd, and other function properties • Cross-Lesson: 8.01 Exponentials & Logarithms

Determine whether $f(x) = x^3 - x$ is even, odd, or neither.

$f(-x) = -x^3 + x = -(x^3 - x) = -f(x)$, so f is odd.

20 [Challenging] • Target: A well-designed word problem: modeling a car rental cost • Cross-Lesson: 1.01 Heart of Algebra

This question has three parts. A rental car company models the cost of a rental as $C(d) = 45d + 20$ dollars, where d is the number of days rented. (a) Find $C(5)$, and interpret its meaning in context. (b) Define $P(d) = C(d) - 200$, representing profit relative to a fixed cost of 200 dollars per rental period. Write $P(d)$ explicitly and find $P(5)$. (c) Find the inverse function $C^{-1}(x)$, and use it to determine how many days someone rented the car if they were charged 335 dollars.

(a) $C(5) = 45(5) + 20 = 245$. Renting the car for 5 days costs 245 dollars total. (b) $P(d) = 45d + 20 - 200 = 45d - 180$. $P(5) = 45(5) - 180 = 225 - 180 = 45$ dollars. (c) From $x = 45d + 20$, solve for d: $d = \frac{x - 20}{45}$, so $C^{-1}(x) = \frac{x - 20}{45}$. For $x = 335$: $d = \frac{335 - 20}{45} = \frac{315}{45} = 7$ days.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Evaluating functions

If $f(x) = -4x^2 + 3x - 4$, find $f(7)$.

a. -177

b. -179 ✓

c. -181

d. -11

$f(7) = -4(7)^2 + 21 - 4 = -179$.

22 [Foundational] • Target: Domain and range

If $f(x) = 3x^2 - 4x + 5$, find $f(6)$.

a. 87

b. 89 ✓

c. 91

d. -1

$f(6) = 3(6)^2 - 24 + 5 = 89$.

23 [Foundational] • Target: Composition of functions

If $f(x) = x^2 - 9x$, find $f(8)$.

a. -64

b. -6

c. -8 ✓

d. -10

$f(8) = (8)^2 - 72 = -8$.

24 [Foundational] • Target: Inverse functions

If $f(x) = -4x^2 + 8x - 8$, find $f(4)$.

a. -42

b. -38

c. 8

d. -40 ✓

$f(4) = -4(4)^2 + 32 - 8 = -40$.

25 [Foundational] • Target: Transformations of graphs

If $f(x) = -4x^2 - 7x + 3$, find $f(8)$.

a. -309 ✓

b. -307

c. -311

d. -85

$f(8) = -4(8)^2 - 56 + 3 = -309$.

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26 [Foundational] • Target: Piecewise functions

If $f(x) = -2x^2 - 3x + 1$, find $f(6)$.

a. -29

b. -87

c. -89 ✓

d. -91

$f(6) = -2(6)^2 - 18 + 1 = -89$.

27 [Foundational] • Target: Reading function graphs

If $f(x) = -4x^2 + 3x + 7$, find $f(8)$.

a. -223

b. -227

c. -1

d. -225 ✓

$f(8) = -4(8)^2 + 24 + 7 = -225$.

28 [Foundational] • Target: Even, odd, and other function properties

If $f(x) = x - 2$ and $g(x) = 4x^2 + 3$, find $(f \circ g)(-6)$.

a. 148

b. 147

c. 145 ✓

d. 143

$g(-6) = 4(-6)^2 + 3 = 147$. $f(147) = (147) - 2 = 145$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: modeling a car rental cost

If $f(x) = 4x - 3$ and $g(x) = 2x^2 + 2$, find $(f \circ g)(0)$.

a. 5 ✓

b. 3

c. 7

d. 2

$g(0) = 2(0)^2 + 2 = 2$. $f(2) = 4(2) - 3 = 5$.

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30 [Intermediate] • Target: Function Notation & Operations

If $f(x) = -3x - 9$ and $g(x) = -4x^2 + 2$, find $(f \circ g)(-3)$.

a. 91

b. -34

c. 95

d. 93 ✓

$g(-3) = -4(-3)^2 + 2 = -34$. $f(-34) = -3(-34) - 9 = 93$.

31 [Intermediate] • Target: Composite Functions & Inverses

If $f(x) = -4x + 3$ and $g(x) = -2x^2 - 5$, find $(f \circ g)(-1)$.

a. 29

b. 31 ✓

c. 33

d. -7

$g(-1) = -2(-1)^2 - 5 = -7$. $f(-7) = -4(-7) + 3 = 31$.

32 [Intermediate] • Target: Graph Transformations

If $f(x) = -x + 3$ and $g(x) = 2x^2 + 3$, find $(f \circ g)(-2)$.

a. -8 ✓

b. -10

c. 11

d. -6

$g(-2) = 2(-2)^2 + 3 = 11$. $f(11) = -(11) + 3 = -8$.

33 [Intermediate] • Target: Rational Graphs & Asymptotes

If $f(x) = 3x - 4$ and $g(x) = -2x^2 - 2$, find $(f \circ g)(1)$.

a. -16 ✓

b. -18

c. -4

d. -14

$g(1) = -2(1)^2 - 2 = -4$. $f(-4) = 3(-4) - 4 = -16$.

34 [Intermediate] • Target: Evaluating functions

If $f(x) = -x - 3$ and $g(x) = 3x^2$, find $(f \circ g)(0)$.

a. -1

b. -5

c. 0

d. -3 ✓

$g(0) = 3(0)^2 = 0$. $f(0) = -(0) - 3 = -3$.

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35 [Intermediate] • Target: Domain and range

Find the inverse of $f(x) = 3x + 4$.

a. $\frac{x-4}{-3}$

b. $\frac{x-4}{3}$ ✓

c. $\frac{x+4}{3}$

d. $3x + 4$

Swap x and y in $y = 3x + 4$: $x = 3y + 4$, so $y = \frac{x-4}{3}$. $f^{-1}(x) = \frac{x-4}{3}$.

36 [Intermediate] • Target: Composition of functions

Find the inverse of $f(x) = 4x + 9$.

a. $\frac{x-9}{-4}$

b. $\frac{x+9}{4}$

c. $\frac{x-9}{4}$ ✓

d. $4x + 9$

Swap x and y in $y = 4x + 9$: $x = 4y + 9$, so $y = \frac{x-9}{4}$. $f^{-1}(x) = \frac{x-9}{4}$.

37 [Intermediate] • Target: Inverse functions

Determine whether $f(x) = 2x^4 + 7x^2$ is even, odd, or neither.

a. both even and odd

b. neither

c. odd

d. **even** ✓

$f(-x) = 2(-x)^4 + 7(-x)^2 = f(x)$, so f is even.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Transformations of graphs • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Determine whether $f(x) = -x^3 + 2$ is even, odd, or neither.

a. both even and odd

b. even

c. **neither** ✓

d. odd

$f(-x) = -(-x)^3 + 2$, which equals neither $f(x)$ nor $-f(x)$, so f is neither even nor odd.

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39 [Challenging] • Target: Piecewise functions • Cross-Lesson: 8.01 Exponentials & Logarithms

A function is defined by $f(x) = -x$ for $x < -3$ and $f(x) = -5x - 5$ for $x \geq -3$. Find $f(-4)$ and $f(2)$.

a. $6, -15$

b. $4, -17$

c. $4, -15$ ✓

d. $-15, 4$

$f(-4) = -(-4) = 4$ (using the first piece, since $-4 < -3$). $f(2) = -5(2) - 5 = -15$ (using the second piece, since $2 \geq -3$).

40 [Challenging] • Target: Reading function graphs • Cross-Lesson: 1.01 Heart of Algebra

A function is defined by $f(x) = 2x + 1$ for $x < 0$ and $f(x) = -3x - 6$ for $x \geq 0$. Find $f(-5)$ and $f(3)$.

a. $-9, -15$ ✓

b. $-9, -17$

c. $-7, -15$

d. $-15, -9$

$f(-5) = 2(-5) + 1 = -9$ (using the first piece, since $-5 < 0$). $f(3) = -3(3) - 6 = -15$ (using the second piece, since $3 \geq 0$).

41 [Challenging] • Target: Function Notation & Operations • Cross-Lesson: 2.01 Quadratics & Polynomials

If $f(x) = 4x - 5$ and $g(x) = 2x^2 + 1$, what is $(f + g)(3)$?

a. 24

b. 26 ✓

c. 28

d. 30

$f(3) = 4(3) - 5 = 7$. $g(3) = 2(3^2) + 1 = 18 + 1 = 19$. $(f + g)(3) = 7 + 19 = 26$.

42 [Challenging] • Target: Function Notation & Operations • Cross-Lesson: 4.01 Ratios, Rates & Data

What is the domain of the function $h(x) = \frac{\sqrt{x-4}}{x-7}$?

a. $[4, 7) \cup (7, \infty)$ ✓

b. $[4, \infty)$

c. $(4, 7) \cup (7, \infty)$

d. $x \neq 7$

For the square root, $x - 4 \geq 0 \Rightarrow x \geq 4$. Denominator cannot be zero: $x \neq 7$. Domain is $[4, 7) \cup (7, \infty)$.

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43 [Challenging] • Target: Composite Functions & Inverses • Cross-Lesson: 5.01 Statistics & Probability

If $f(f^{-1}(x)) = x$ for all x, and $f(4) = 9$, what is $f^{-1}(9)$?

a. 4 ✓

b. $\frac{1}{4}$

c. $\frac{1}{9}$

d. 9

By definition of the inverse function, if $f(a) = b$, then $f^{-1}(b) = a$. Since $f(4) = 9$, $f^{-1}(9) = 4$.

44 [Challenging] • Target: Graph Transformations • Cross-Lesson: 6.01 Geometry

The graph of $y = x^2$ is vertically stretched by a factor of 3 and reflected across the x-axis. What is its equation?

a. $y = -3x^2$ ✓

b. $y = 3(-x)^2$

c. $y = -\frac{1}{3}x^2$

d. $y = -(3x)^2$

Vertical stretch by 3 multiplies the function by 3: $3x^2$. Reflection across the x-axis negates the function: $-3x^2$.

45 [Challenging] • Target: Rational Graphs & Asymptotes • Cross-Lesson: 7.01 Trigonometry & Additional Topics

What is the vertical asymptote of $f(x) = \frac{3x+1}{x-5}$?

a. $x = 5$ ✓

b. $x = -5$

c. $y = 3$

d. $y = 5$

A vertical asymptote occurs where the denominator is zero (and numerator is non-zero): $x - 5 = 0 \Rightarrow x = 5$.

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